

Reactive splicing compatibilization of immiscible polymer blends: Compatibilizer synthesis in the melt state and compatibilizer architecture effects

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ABSTRACT

Reactive compatibilization is an effective method to improve the compatibility of immiscible polymer blends. The in-situ-formed graft/block copolymers should be thermodynamically located at the interface to bridge the neighboring phases. Unfortunately, they are often pulled out of the interface because of the asymmetric molecular structures. Here, we propose a new strategy to compatibilize immiscible polymer blends called reactive splicing compatibilization. Poly (styrene-co-glycidyl methacrylate) (SG) with high glycidyl methacrylate (GMA) content is used as the reactive backbone chain. Both poly (L-lactic acid) (PLLA) and poly (butylene adipate-co-terephthalate) (PBAT) molecular chains can be easily grafted onto the SG main backbone by the reaction of the terminal carboxylic acid groups with the epoxide groups in the melt. Direct melt blending SG with PLLA and/or PBAT leads to reactive compatibilizers with various molecular architectures, which are used for subsequent compatibilization of PLLA/PBAT blends. It was found that the compatibilizer architecture significantly affects the compatibilization efficiency. The best compatibilization occurs with a reactive compatibilizer of SG-g-PBAT with the SG to PBAT weight ratio of 1:2. The compatibilized PLLA/PBAT (50/50 w/w) blend has precise co-continuous phase morphology with very few micelles. The blend exhibits extremely high Charpy notched impact strength of 90.2 kJ/m² and tensile strength of 33.3 MPa. The principle of the facile splicing compatibilization strategy is highly applicable to many other immiscible blends with both components containing reactive groups.

1. Introduction

Polymer blending is one of the most feasible strategies to fabricate new materials with a favorable performance combination of the components [1–6]. Because of the limited entropy contributions of high molecular polymers during blending, most polymer blends are incompatible and the formed blends exhibit even poorer properties than the components [7–12]. Addition of pre-made block/graft co-polymers or reactive compatibilizers to the blends is necessary to decrease the interfacial tension and suppress phase coalescence [13–18]. Widely used reactive linear compatibilizers have a backbone that is miscible with one component and reactive groups at the chain end or randomly distributed along the backbone that react with the complementary groups of the other component during melt blending [19–24]. Therefore, the in-situ-formed block/graft copolymer compatibilizers locate at the

interface to bridge the neighboring phases [25–30]. Obviously, the compatibilization efficiency is greatly dependent on the molecular structure of the starting reactive compatibilizers [31]. Charoensir-isonboon et al. found that in-situ-formed graft polymers are more stable at the interface than in-situ-formed block polymers under the shear condition [32]. Kim et al. reported direct evidence that the interfacial morphology of the blend significantly depends on the chain architecture of the in-situ-formed graft or block copolymers [33]. Jeon et al. also suggested that graft polymers are more efficient than block polymers in the reactive case because graft polymers show greater ability to reduce the interfacial tension [34].

Unfortunately, there are also some disadvantages, such as unknown compatibilizer composition, incompletely complex reaction conversion between components, and limited reaction types for the reactive compatibilization [35,36]. One of the most importation limitations is the fact

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that the in-situ-formed graft copolymers are not always located at the interface to act as an efficient compatibilizer. They are usually pulled out of the interface and form micelles in one of the two phases under shearing during melt blending at high temperature. Macosko [35] and Inoue [37] have systematically investigated the formation mechanism of the micelles in reactive compatibilized polymer blends. They attributed migration of the in-situ-formed graft copolymer to the asymmetric molecular structure and unbalanced interactions between the graft copolymer and both components. This leads to “pulling in” or “pulling out” of the graft copolymer from the reaction location, as shown in Fig. 1(a). To overcome the unbalanced interactions of one-side-grafted copolymers and achieve high efficiency of reactive compatibilizers, we recently proposed the double-comb copolymer compatibilization strategy for immiscible polymer blends using reactive grafted copolymers (RGCs) as the compatibilizer [38,39]. The poly(methyl methacrylate-co-glycidyl methacrylate)-graft-poly(methyl methacrylate) (P((MMA-co-GMA)-g-MMA)) RGC was synthesized by the grafting through method and then used as a reactive compatibilizer for immiscible poly(vinylidene fluoride)/poly(L-lactic acid) (PVDF/PLLA) blends [40]. The terminal carboxyl acid groups of PLLA reacted with the epoxy groups of the RGC and double-grafted compatibilizers formed in situ at the interface, as shown in Fig. 1(b). Compared with one side-grafted compatibilizers, double-grafted compatibilizers are more stable at the interface and show a significantly enhanced compatibilization effect [38,39].

Although RGCs show high compatibilization efficiency for immiscible polymer blends, the RGC strategy still suffers from several limitations. RGC synthesis using macromers is difficult and requires at least two synthesis steps. The molecular structure of the RGC is also limited because of the different copolymerization rates of macromers and small molecule monomers. In addition, the synthesized RGC is usually only applicable for a specific blend during compatibilization. We have developed a creative strategy to avoid the complicated synthesis process of the RGC for polymer blends with both components containing reactive groups called splicing compatibilization (Fig. 2). Reactive linear copolymers can be used as efficient compatibilizers for A/B blends. Both polymer A and B contain reactive groups with reactive linear copolymers. Therefore, both A and B can be spliced or grafted onto the reactive linear copolymers to achieve double-grafted compatibilizers. We expect that the formed double-grafted copolymers can effectively compatibilize A/B blends. Formation of double-grafted copolymers is crucial for high compatibilization efficiency. In this study, we used

PLLA/PBAT blends as the model system and poly(styrene-co-glycidyl methacrylate) (SG) as the reactive linear compatibilizer. Both PLLA and PBAT contain reactive carboxylic acid groups that can react with the epoxide groups of SG in the melt state. PLLA/PBAT blends are typical immiscible blends, and they have been extensively investigated as completely biodegradable polymers [41–44]. Both PLLA and PBAT can be attached/spliced to the SG main chain, but the grafting kinetics is different. Therefore, the effect of the splicing sequence of PLLA and PBAT onto SG on compatibilization was investigated. It was found that compatibilizers with double-spliced PLLA and PBAT chains exhibit the best compatibilization efficiency for the blends.

2. Experimental section

2.1. Materials

Poly L-lactide (PLLA, 3001D, $M_n = 8.9 \times 10^4$, $M_w/M_n = 1.8$) was supplied from NatureWorks (USA). PBAT (Biocose 2003) was purchased from Yifan Fuxin Co., Ltd in China. SG (glycidyl methacrylate copolymerization ratio 45 wt%) was kindly provided by Eco New Material Co., Ltd in China.

2.1.1. RGC preparation

PLLA, PBAT, and SG were dried at 60 °C for 8 h in a vacuum oven before processing. PLLA, PBAT, and SG were then mixed at 190 °C in a mixer (Haake PolyLab QC) with a rotation speed of 50 rpm until the torque did not increase. The molar equivalent of epoxy groups in SG should be much higher than the terminal carboxyl groups in PBAT and PLLA to ensure that both PLLA and PBAT graft onto SG and the RGC is reactive. Moreover, the structure of the RGC can be regulated by adjusting the feed ratio of PLLA, PBAT, and SG. We define PBAT and PLLA side-chain-grafted SG as LxSyBz, where L is PLLA, S is SG, B is PBAT, and x, y, and z are the feed ratios of PLLA, SG, and PBAT by weight, respectively. The codes and feed ratios of the reactive graft compatibilizers are given in Table 1.

2.2. PLA/PBAT blend preparation

The PLLA/PBAT blends with compatibilizers having different architectures were prepared in a mixer (Haake PolyLab QC) at a rotation speed of 50 rpm and 190 °C. The compatibilized blends had a constant

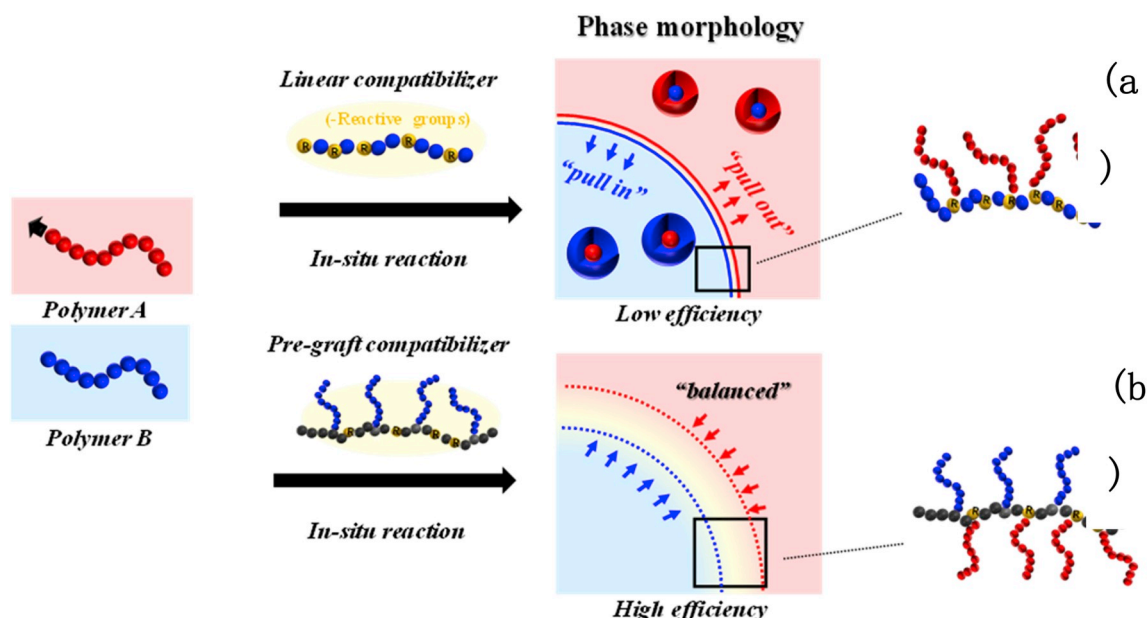


Fig. 1. Schematic diagrams of the compatibilization mechanisms for (a) a reactive linear compatibilizer and (b) a reactive grafted compatibilizer.

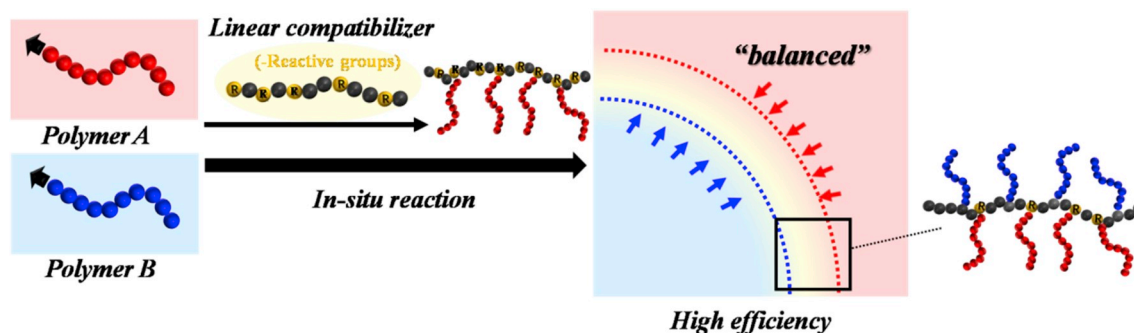


Fig. 2. Schematic diagram of the reactive splicing compatibilization using reactive linear compatibilizers.

Table 1

Preparation of the reactive grafted compatibilizers by melt blending with various PLLA:SG:PBAT feed ratios.

Compatibilizer Code	Formula of the compatibilizers	PBAT content in compatibilizer (wt %)	PLLA content in compatibilizer (wt %)
L2S1	PLLA 40g; SG 20g	–	66.6
S1B2	SG 20g; PBAT 40g	66.6	–
L1S1B1	PLLA 20g; SG 20g; PBAT 20g	33.3	33.3
S1B1	SG 30g; PBAT 30g	50	–
S1B3	SG 15g; PBAT 45g	75	–
S1B4	SG 12g; PBAT 48g	80	–

PLLA/PBAT/SG component ratio of 50/50/3. All of the prepared samples were hot-pressed into sheets at 200 °C under a pressure of 10 MPa and cooled to 30 °C with water. The obtained sheets were used for subsequent characterization.

2.3. Characterization

Mechanical Tests. The tensile and impact samples were prepared with a micro-injection molding machine (Haake Minijet Pro). A universal test machine (INSTRON-5966) was used to determine the tensile properties. The dumbbell-shaped specimens prepared with the injection molding machine were strained at a constant crosshead speed of 5 mm/min at room temperature. The notched Charpy impact test was performed according to the GB/T 16420-1996 standard with an impact tester (SS-3700) and a pendulum energy of 4 J.

Scanning Electron Microscopy (SEM). A field emission scanning electron microscope (Hitachi S-4800) was used to determine the morphologies of the blends. All of the blends were fractured in liquid nitrogen. Before morphology characterization, the fractured surfaces were treated with gold.

Transmission Electron Microscopy (TEM). The morphologies of the blends were characterized by TEM (Hitachi H-800). The specimens were sliced to 70–90 nm and stained with RuO₄ for 4 h for better contrast.

Differential Scanning Calorimetry (DSC). DSC (Q 2000) was performed at a heating/cooling rate of 3 °C/min in a nitrogen atmosphere, and the testing temperature range was –50 to 200 °C.

Fourier Transform Infrared (FTIR) Spectroscopy. Before the tests, all of the samples were pressed into sheets and their FTIR spectra were recorded with a Germany Bruker VERTEX 70v spectrometer. All of the sheets were scanned 64 times from 400 to 4000 cm^{–1} at a resolution of 2 cm^{–1} by attenuated total reflection.

Thermogravimetric Analysis (TGA). The TGA measurements were performed with a TGA-Q500 analyzer (TA Instruments) in a nitrogen atmosphere. The specimens were heated from 30 to 650 °C at a rate of 10 °C/min.

Dynamic Mechanical Analysis (DMA). Samples were molded to dimensions of 8 mm × 6.30 mm × 0.50 mm (length, width, and thickness)

and then tested with a DMA Q800 dynamic mechanical analyzer with an amplitude of 5 μm, a frequency of 5 Hz, and a heating rate of 3 °C/min from –50 to 150 °C.

3. Results

3.1. Synthesis of RGCs with different structures in the melt

A series of RGCs with different structures were obtained by direct melt blending of SG with PLLA and/or PBAT. Compatibilizers with different architectures were prepared by varying the weight ratio of SG to PBAT/PLLA (Table 1). It should be noted that SG contains high epoxide group content and all of PBAT and PLLA can be fully grafted onto SG with remaining unreacted epoxide groups on the SG chains. To confirm successful synthesis of the different reactive grafted compatibilizers, the prepared samples were all purified using toluene and acetonitrile to eliminate possible un-grafted species. The FTIR spectra of neat PLLA, PBAT, SG, and L2S1, S1B2, and L1S1B1 after purification are shown in Fig. 3(a). The infrared absorption peaks of L1S1B1 at 1715, 1708, and 910 cm^{–1} represent the carbonyl groups of PLLA, carbonyl groups of PBAT, and epoxy groups of SG, respectively. It is clear that PLLA and PBAT can be easily grafted onto SG, and PLLA-grafted-SG-grafted-PBAT (PLLA-g-SG-g-PBAT), PLLA-grafted-SG (PLLA-g-SG), and PBAT-grafted-SG (PBAT-g-SG) can be prepared by melt blending. The presence of the absorption peak at 910 cm^{–1} confirms that all of the graft copolymers contain epoxide groups for the following reaction during melt mixing. Formation of PLLA-g-SG-g-PBAT, PLLA-g-SG, and PBAT-g-SG was also confirmed by the DSC measurements (Fig. 3(b)). Purified PLLA-g-SG-g-PBAT had two glass transitions. The glass transition at –24 °C is assigned to the segment motion of grafted PBAT, whereas the other glass transition located at 58 °C is assigned to that of grafted PLLA. The glass transition of SG may be overlapped by that of PLLA owing to their close glass transition temperatures, so only one glass transition was observed for PLLA-g-SG. However, PBAT-g-SG also had two glass transitions corresponding to PBAT and SG. The derivative thermogravimetry (DTG) curves of pure PLLA, PBAT, and SG, and purified L2S1, S1B2, and L1S1B1 are shown in Fig. 3(c). The curves for pure PLLA, PBAT, and SG only exhibit one peak, while that for PLLA-g-SG-g-PBAT exhibits two peaks. The first peak corresponds to the degradation of PLLA, the second peak is assigned to that of PBAT, and the peak of SG is masked by those of PLLA and PBAT. The DTG curves of PBAT-g-SG and PLLA-g-SG also show the characteristics of grafted PBAT and PLLA, respectively. Moreover, as for PLLA-g-SG-g-PBAT, microphase separation can be observed in TEM and the nanodomains (PLLA) with sizes of 80–100 nm disperse in matrix (PBAT) in Fig. S1. It is concluded that PLLA and PBAT react with SG and form reactive grafted compatibilizers with both grafts and remaining epoxide groups on the main chains.

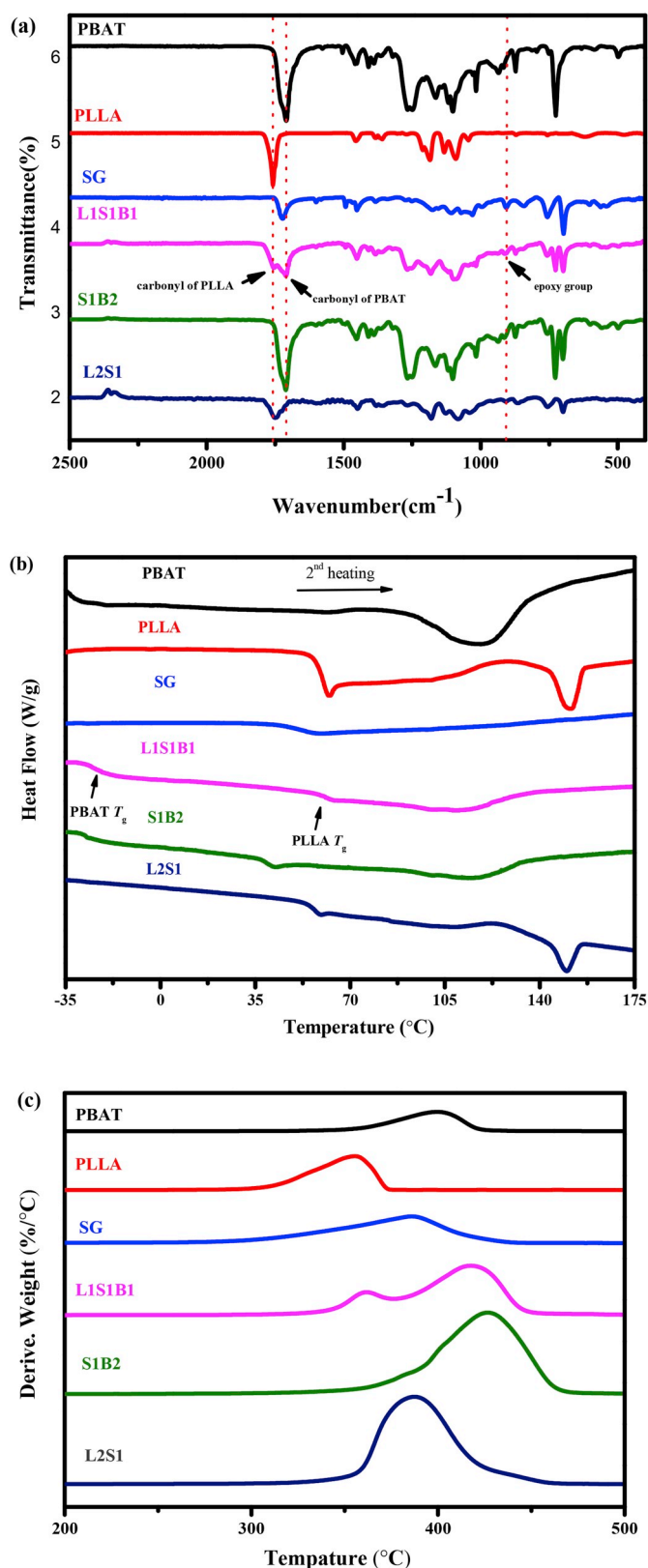


Fig. 3. Characterization of pure PLLA, PBAT, SG, L1S1, S1B2, and L1S1B1: (a) FT-IR spectra, (b) DSC curves, and (c) DTG curves.

3.2. Reactive compatibilization of PLLA/PBAT blends

3.2.1. Compatibilizer architecture effects

The melt-prepared RGCs were used as compatibilizers for PLLA/

PBAT blends. SEM images of PLLA/PBAT (50/50 w/w) blends with and without reactive compatibilizers with different architectures are shown in Fig. 4. For the binary blend without a compatibilizer (Fig. 4(a)), the PLLA phase was dispersed in the PBAT matrix owing to the higher melt viscosity of PLLA than PBAT. There were large gaps between the PLLA dispersed phase and the PBAT matrix, indicating the thermodynamic immiscible behavior of the PLLA/PBAT blend. Addition of 3 wt% SG markedly decreased the size of PLLA dispersed phase (Fig. 4(b)). In addition, interface adhesion between PLLA and PBAT was enhanced. This means that SG has a favorable compatibilization effect on the PLLA/PBAT blend, which is ascribed to the possible reactions between the epoxide groups of SG and the carboxylic acid groups of both PBAT and PLLA. The L2S1 compatibilized blend had a sea-island structure (Fig. 4(c)). However, the PLLA phase was larger and the interface between PLLA and PBAT was poorer compared with the SG compatibilized blend. Clearly, L2S1 is not suitable for compatibilization of the PLLA/PBAT (50/50 w/w) blend. The compatibilization effect of the reactive double-grafted copolymer (L1S1B1) was also investigated. The PLLA/PBAT/L1S1B1 blend also had a sea-island structure (Fig. 4(d)). For the reactive single-grafted copolymer S1B2, the PLLA/PBAT with S1B2 blend exhibited a completely different microstructure from the other blends. Incorporation of S1B2 changed the morphology from a sea-island structure to a co-continuous structure (Fig. 4(e)). Moreover, the interface between the two phases was significantly enhanced. Considering the above-mentioned facts, it is clear that reactive single-grafted S1B2 has an excellent compatibilization effect on the PLLA/PBAT (50/50 w/w) blend.

TEM images of PLLA/PBAT (50/50 w/w) blends with and without reactive compatibilizers with different architectures are shown in Fig. 5. PLLA cannot be stained by RuO₄ and it appears white in the TEM images. In contrast, PBAT with benzene rings is easily stained by RuO₄ and appears black [43]. The morphologies obtained by the TEM measurements are consistent with the SEM results. The binary PBAT/PLLA blend without a compatibilizer showed a typical sea-island morphology with very broad PLLA phase size distribution (Fig. 5(a)). There were many very big PLLA domains, but they cannot be observed in Fig. 5(a) owing to the large magnification of the TEM technique. Incorporation of SG, L2S1, L1S1B1, and S1B2 resulted in significantly different phase morphologies. Sea-island morphologies with irregular interfacial shapes were observed for the blends compatibilized by SG, L2S1, and L1S1B1 (Fig. 5(b)–(d)). Moreover, numerous micelles were present in the PLLA domains of these compatibilized blends, but the micellar number (concentration) in the PLLA domains was different for the three compatibilized blends. The L2S1 compatibilized PLLA/PBAT blend contained the most micelles in the PLLA phase. The blend compatibilized by the S1B2 compatibilizer had significantly different phase morphologies (Fig. 5(e)). The blend had a typical co-continuous structure and the curvature of the PLLA/PBAT interface was small. Such a structure is beneficial for synergistic property combination of the two phases. Differing from the blends compatibilized with the other three types of compatibilizer, there were very few micelles in the PLLA phase. This means that S1B2 has the best compatibilization effect on the PLLA/PBAT blend.

The morphology of the polymer blends is a crucial factor that influences the mechanical properties. The notched impact strength and stress-strain curves of PLLA/PBAT blends without and with different reactive compatibilizers are shown in Fig. 6(a) and (b), respectively. The mechanical properties of the binary PBAT/PLLA blend without a compatibilizer were poor, with elongation at break of about 12% and notched impact strength of 4.5 kJ/m². This is because of the weak interface between the PBAT and PLLA phases, as evidenced by the SEM morphology in Fig. 4(a). With addition of 3 wt% SG, the elongation at break and tensile strength of the blend increased to 80% and 27.6 MPa, respectively. The notched impact strength of the SG compatibilized blend was 19.5 kJ/m², which is about 4.5 times higher than that of the binary blend without a compatibilizer. This means that the linear reactive SG polymer enhances the compatibility between PLLA and

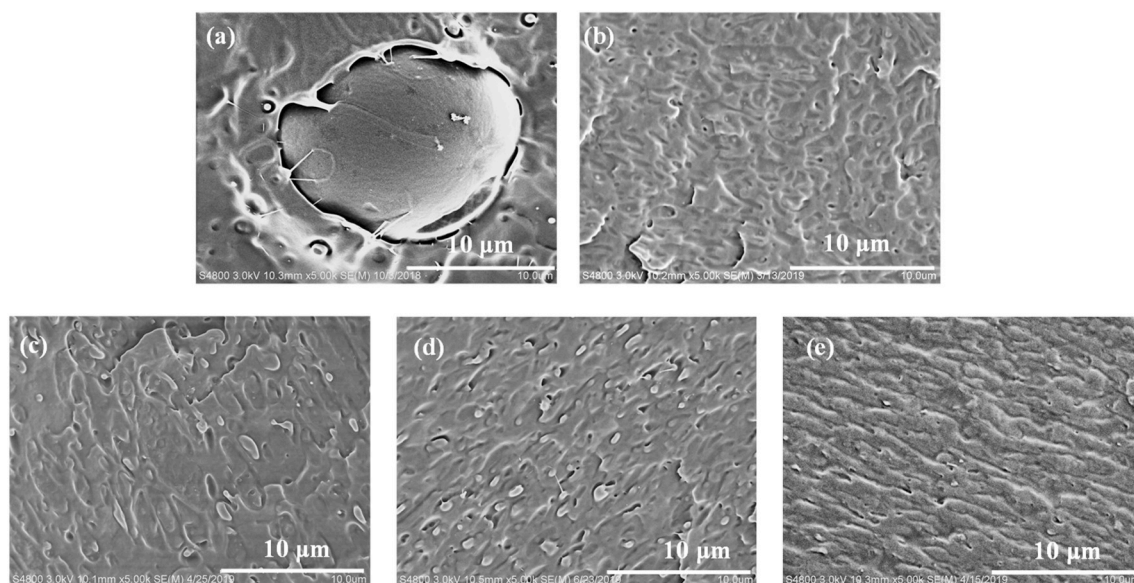


Fig. 4. SEM images of the PLLA/PBAT blends (a) without compatibilizer, (b) compatibilized by SG, (c) compatibilized by L2S1, (d) compatibilized by L1S1B1, and (e) compatibilized by S1B2 blends.

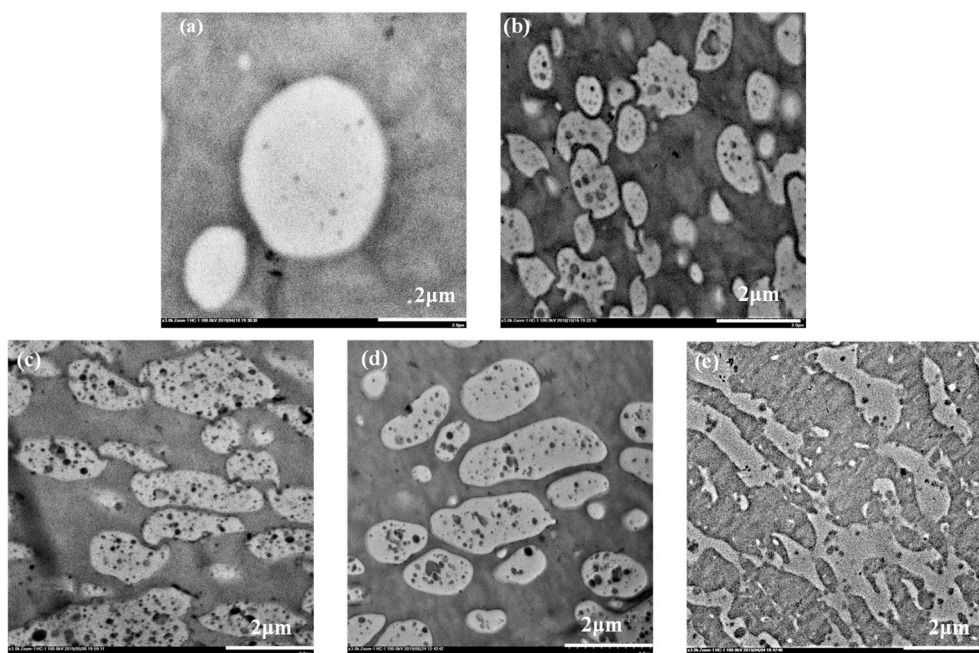


Fig. 5. TEM images of the PLLA/PBAT blends (a) without compatibilizer, (b) compatibilized by SG, (c) compatibilized by L2S1, (d) compatibilized by L1S1B1, and (e) compatibilized by S1B2 blends.

PBAT. The L2S1 compatibilized blend had similar mechanical properties, with elongation at break of 76% and notched impact strength of 18.7 kJ/m^2 . The elongation at break and notched impact strength of the blend containing the L1S1B1 compatibilizer with both pre-grafted PBAT and pre-grafted PLLA were slightly enhanced compared with the blend containing the L2S1 compatibilizer. The S1B2 compatibilized blend had significantly different mechanical properties, with elongation at break and tensile strength of 200% and 33.3 MPa, respectively. Furthermore, the notched impact strength of the blend was 90.2 kJ/m^2 , which is 20 times higher than that of the PLLA/PBAT blend without a compatibilizer. Photographs of impact samples of the PLLA/PBAT blends are exhibited in Fig. S2. The S1B2 compatibilized blends were not fully broken after the notched impact test and very limit broken zone can be

observed. However, the other blends were fully fractured. It should be noted that the mechanical properties of the S1B2-compatibilized PBAT/PLLA blend are adequate for many engineering applications due to the superior toughness and the high tensile strength. [45]. The elongation at break, tensile strength, and notched impact strength of the PLLA/PBAT/L1S1B1 blend are close to those of the blends with SG and L2S1, but clearly lower than those of the PBAT/PLLA/S1B2 blend. This indicates that S1B2 has the best compatibilization effect on the PBAT/PLLA (50/50 w/w) blend, which is in good agreement with morphologies in Fig. 4.

The variations in the glass-transition temperatures (T_g) of the components of the PLLA/PBAT blends can also reflect the compatibilization effects of the reactive compatibilizers with different architectures. The $\tan \delta$ curves from DMA measurements of PBAT/PLLA blends with and

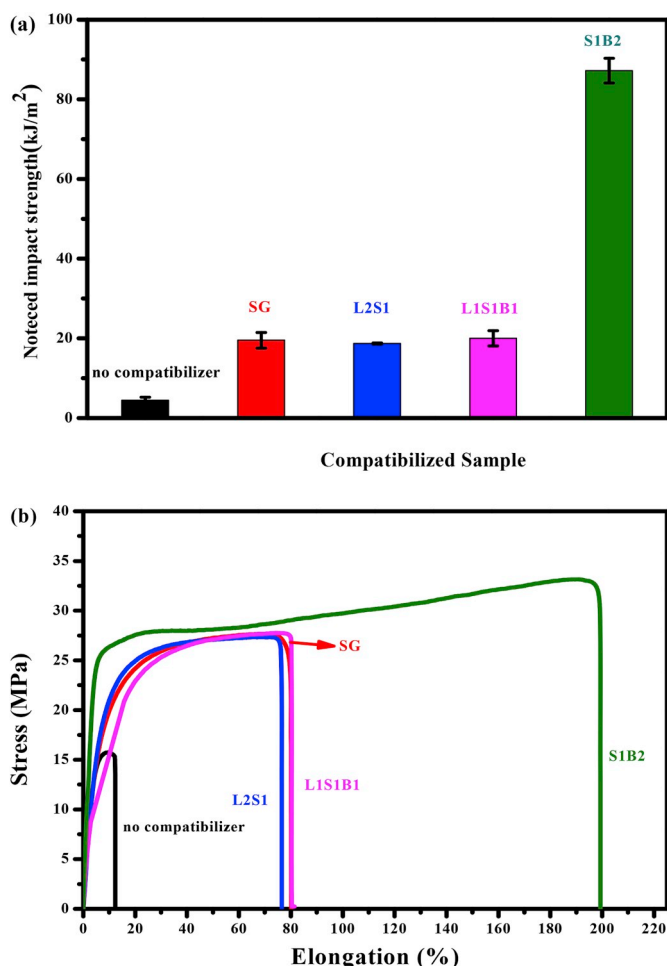


Fig. 6. (a) Notched impact strength and (b) tensile strain–stress curves of the PLLA/PBAT blends compatibilized by the indicated compatibilizers.

without SG and different reactive grafted compatibilizers are shown in Fig. 7. The T_g value of PLLA shifted to lower temperature with addition of reactive compatibilizers to the PLLA/PBAT blend, but T_g of PBAT shifted to higher temperature for the S1B2 compatibilized blend. The glass transition temperature difference between PLLA and PBAT (ΔT_g)

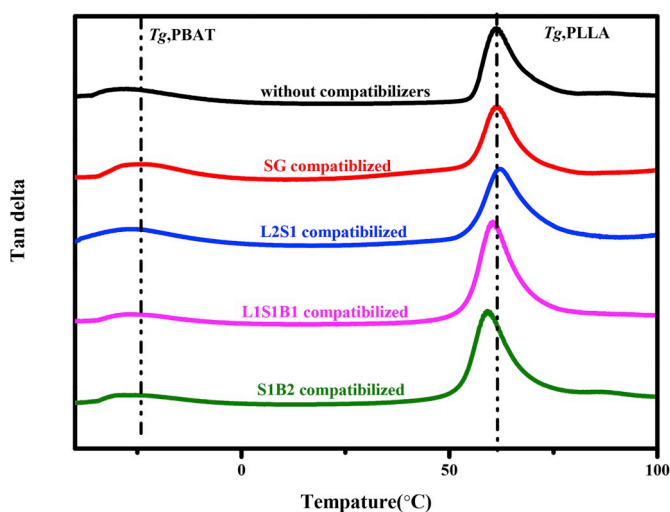


Fig. 7. Tan δ curves from DMA measurements of the PLLA/PBAT blends with and without a compatibilizer.

decreased with addition of reactive compatibilizers to the PBAT/PLLA blend (Table 2). For the PLLA/PBAT/S1B2 blend, $\Delta T_g = 84.5^\circ\text{C}$, which is much lower than that of 88.6°C for the blend without a compatibilizer, indicating greatly improved compatibility between PLLA and PBAT. This again indicates that S1B2 has the best compatibilization effect on the PLLA/PBAT blend.

3.2.2. Optimization of the structure of the reactive grafted compatibilizer

Among the compatibilizers considered, the reactive S1B2 compatibilizer shows the best compatibilization effect for the PLLA/PBAT blend. The structure of PBAT pre-grafted compatibilizer can be optimized by changing the amount of PBAT grafted onto SG. It is therefore interesting to investigate the effect of the PBAT pre-grafting amount on the compatibilization efficiency. It is easy to synthesize reactive S1B n compatibilizers with various PBAT to SG ratios by melt blending. SEM images of PLLA/PBAT (50/50 w/w) blends compatibilized with different S1B n compatibilizers are shown in Fig. 8. With addition of S1B1 to the PLLA/PBAT blend, the morphology was very close to a co-continuous structure (Fig. 8(a)). Incorporation of S1B2 changed the morphology to a definite co-continuous structure with a very good interface (Fig. 8(b)). The S1B3 compatibilized blend also had a co-continuous structure, but it had large phase sizes and weak interfaces (Fig. 8(c)). The blend compatibilized with S1B4 changed to a typical sea-island morphology with a clear gap at the interface (Fig. 8(d)), indicating that too much grafted PBAT results in the location of the reactive grafted compatibilizer in the PBAT (matrix) phase and form the micelles in PBAT matrix. This is confirmed by the TEM images of the S1B4 compatibilized blends, as shown in Fig. 8 (d'). Some gray micelles were observed in PBAT matrix.

The notched impact strength and stress–strain curves of PLLA/PBAT blends with different S1B n compatibilizers are shown in Fig. 9. With addition of S1B1 to the PBAT/PLLA blend, both the notched impact strength (70.2 kJ/m^2) and elongation at break (178.5%) were slightly lower than those of PBAT/PLLA/S1B2 (Fig. 6). However, with addition of the S1B3 compatibilizer, the mechanical properties of the PLLA/PBAT (50/50 w/w) blend deteriorated. The notched impact strength decreased to 52.2 kJ/m^2 and the elongation at break and tensile strength decreased to 138.5% and 28.3 MPa , respectively. The performance of the PLLA/PBAT blend significantly decreased with addition S1B4, with both the notched impact strength (7.7 kJ/m^2) and tensile strength (24.3 MPa) greatly decreasing. The physical properties of the blends with different S1B n type compatibilizers are consistent with the SEM results, with S1B2 showing the best compatibilization effect.

Please note that the morphology of S1B2 compatibilized blends is very stable upon annealing. No phase coarsening and morphology change have been observed after the annealing at 200°C under the static state, as shown in Fig. S4. This again means that S1B2 exhibits excellent compatibilization effects for the PLLA/PBAT blends.

4. Discussion

It is interesting that pre-grafted SG with different architectures had completely different compatibilization effects on the PLLA/PBAT blend. We confirmed that both PLLA and PBAT can be grafted onto SG during the melt blending process. However, PLLA and PBAT are two different types of aliphatic polyester. It is reasonable that the terminal carboxylic acid groups of PLLA and PBAT showed different reactivities with the

Table 2

T_g and ΔT_g values from the DMA curves of the PLLA/PBAT blends.

Sample	PBAT T_g ($^\circ\text{C}$)	PLLA T_g ($^\circ\text{C}$)	ΔT_g ($^\circ\text{C}$)
PLLA/PBAT	−27.2	61.4	88.6
PLLA/PBAT/SG	−25.5	61.4	86.9
PLLA/PBAT/L2S1	−27.0	61.5	88.5
PLLA/PBAT/L1S1B1	−26.2	60.5	86.7
PLLA/PBAT/S1B2	−25.4	59.1	84.5

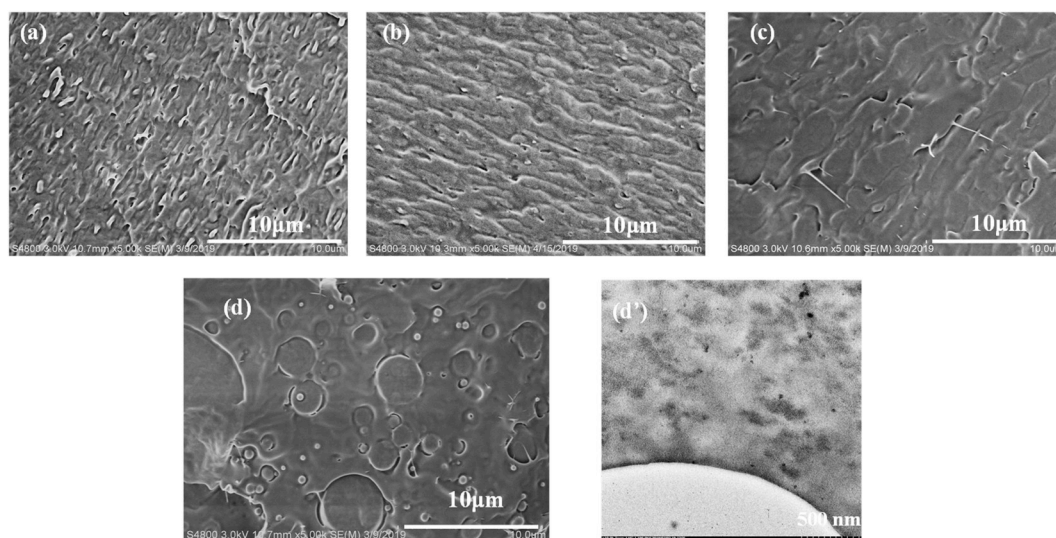


Fig. 8. SEM and TEM images of S1Bn compatibilized PLLA/PBAT blends (a) SEM of S1B1 compatibilized, (b) SEM of S1B2 compatibilized, (c) SEM of S1B3 compatibilized, (d) SEM of S1B4 compatibilized, (d') TEM of S1B4 compatibilized blend.

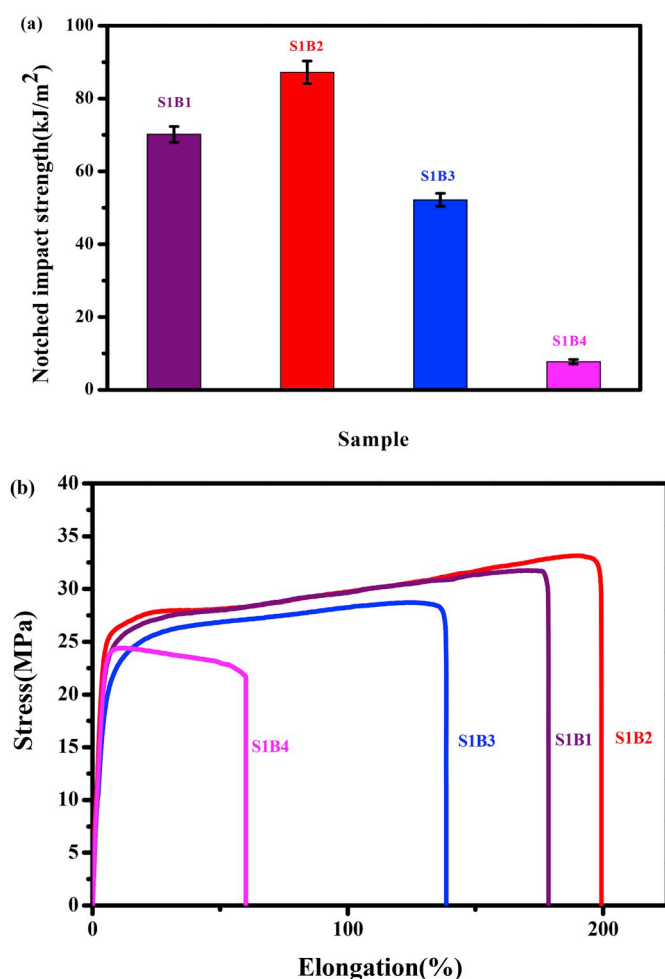


Fig. 9. (a) Notched impact strength and (b) tensile strain–stress curves of the PLLA/PBAT blends compatibilized by the indicated compatibilizers.

epoxide groups on the SG molecular chains. We therefore investigated the difference in the reactivities of PLLA and PBAT with SG. The mixing torque as a function of the mixing time for PLLA/SG and PBAT/SG

blends with the same PLLA/SG and PBAT/SG weight ratios is shown in Fig. 10. The grafting of PLLA and/or PBAT onto SG leads to the increasing in the torque due to the enhanced molecular weight. The time of the ascending torque of the PLLA/SG blend was clearly earlier than that of the PBAT/SG blend. This means that PLLA shows higher reactivity with SG than PBAT under the melt blending conditions.

It is therefore reasonable to understand the compatibilizer molecular architecture effects on the compatibilization efficiency of the immiscible polymer blends. Simultaneous melt mixing of PLLA, PBAT, and SG led to priority grafting of PLLA with SG, so the incorporation SG into the blends leads to the higher splicing possibility of PLLA than PBAT. Therefore, the dominant PLLA grafted copolymers were formed during melt blending and the copolymers were easily pulled into PLLA phase. Numerous micelles were found in the PLLA domains, as shown in Fig. 5 (b) and schematically represented in Fig. 11(a). The reactive L2S1 (reactive PLLA pre-grafted compatibilizer) showed even worse effects on the morphology and properties of the PLLA/PBAT blend compared to the SG copolymers (Fig. 11(b)). The higher reactivity of PLLA leads to the possibly further splicing of PLLA onto the PLLA pre-grafted compatibilizers. Again the PLLA grafted copolymers with less PBAT grafts were also easily to be pulled in the PLLA phase, as evidenced in Fig. 5 (c).

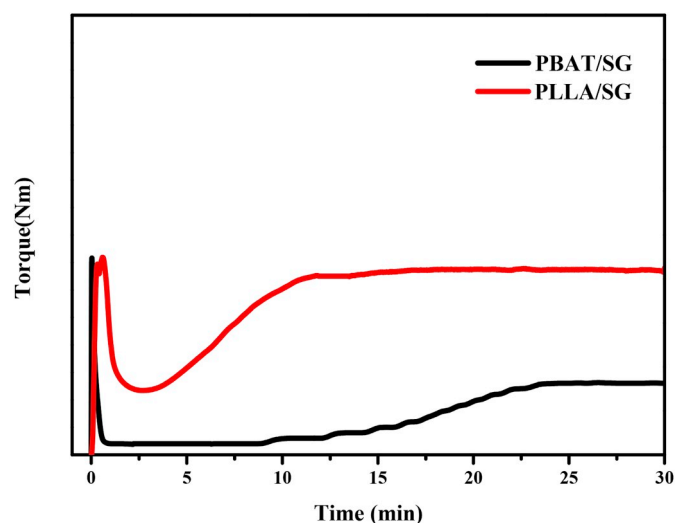


Fig. 10. Melt torque versus time for PLLA/SG and PBAT/SG blends.

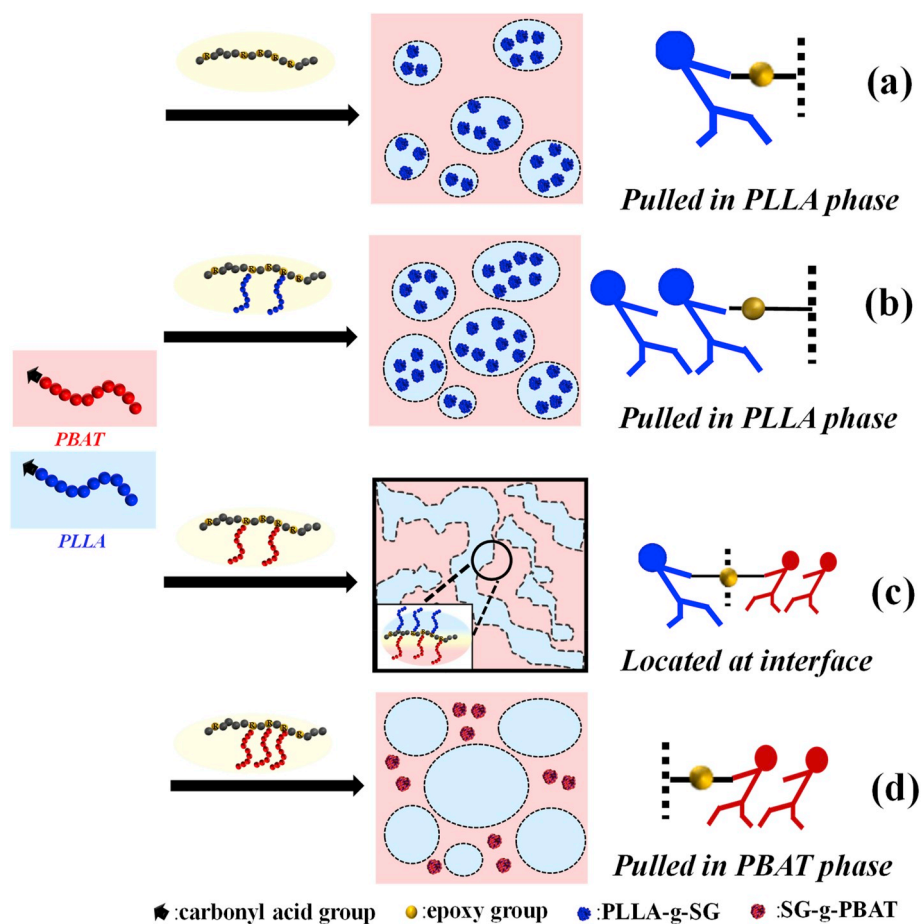


Fig. 11. Schematic diagrams of the compatibilization mechanisms of PLLA/PBAT blends for different reactive compatibilizers.

Introduction of PBAT chains into the PLLA-g-SG molecules slightly enhanced the compatibilization effect for the PLLA/PBAT blend. This is attributed to the slightly balanced interactions between the double grafts and each phase. Therefore, the micelles in the PLLA phase slightly decreased when compatibilized by L1S1B1, as compared with the blend compatibilized by L2S1. However, the compatibilization effect was completely different when only PBAT was first grafted onto SG. The S1B2 copolymer contained both pre-grafted PBAT side chains and remaining epoxide groups. Incorporation of S1B2 into the PLLA/PBAT blend led to subsequent PLLA chain grafting onto the S1B2 copolymer and double-grafted copolymers can be achieved. These double-comb molecules with both PLLA and PBAT side chains can be stably located at the interface, which is attributed to the balanced interactions between the grafts and the molecular chain of each phase, as schematically shown in Fig. 11(c). Both PLLA and PBAT side chains can be confirmed from the FTIR spectra of the remains of the S1B2 compatibilized blends, as evidenced in Fig. S3. They cannot be pulled out of the interface and show a highly efficient compatibilization effect on the PLLA/PBAT blend. Obviously, the strong interface adhesion can transfer stress effectively and the S1B2 compatibilized blends exhibit good mechanical properties. It should be noted that too much pre-grafted PBAT led to decreased PLLA grafting onto, for example, the S1B4 chains, and they can only be located in the PBAT phase, as shown in Fig. 8(d)' and Fig. 11 (d). Therefore, very limited compatibilization effect was observed.

We have demonstrated that the grafting sequence effects on the compatibilization of the immiscible polymer blends. It is considered that the principle is feasible for the compatibilization of many other polymer blends with components containing reactive groups. The two side grafts of the final compatibilizers should have similar affinity with the both phases, so the grafting sequence is critical for the compatibilization. The

two-side-grafted compatibilizers with high compatibilizing efficiency can be prepared by splicing the two side chains onto the backbone and the splicing contents can also be adjusted by the direct melt blending.

5. Conclusion

Both PLLA and PBAT can be easily spliced onto the SG backbone owing to the reaction between the carboxylic acids and epoxide groups. A series of reactive grafted compatibilizers with PLLA and/or PBAT side chains and reactive epoxide groups was prepared by melt blending. The prepared reactive graft copolymers were then used as reactive compatibilizers for immiscible PLLA/PBAT blends. PLLA has higher reactivity with SG than PBAT. Addition of SG and PLLA-g-SG (L2S1) to the blend induced to further PLLA splicing into the SG backbone and the formed graft copolymers were pulled into the PLLA phase. SG-g-PBAT (S1B2) had the best compatibilization effect because of the pre-grafted PBAT chains and the subsequently grafted PLLA chains. Such double-grafted compatibilizers have a good compatibilization effect on the PLLA/PBAT blends. The compatibilized blends exhibit extremely high notched impact strength and high tensile strength. This splicing compatibilization strategy leads to the possibility of both molecular architecture design and production of high-performance polymer blends.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.polymer.2019.121952>.

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